

Chapter

Ratio And Proportion

1. The table shows the values of x and y , where x is proportional to y

x	6	12	N
y	M	18	6

What are the values of M and N ?

- (a) $M = 4, N = 9$
- (b) $M = 9, N = 3$
- (c) $M = 9, N = 4$
- (d) $M = 12, N = 0$

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2. The given table shows the distance covered and the time taken by a train moving at a uniform speed along a straight track.

Distance (in m)	60	90	y
Time (in sec)	2	x	5

The values of x and y are :

- (a) $x = 4, y = 150$
- (b) $x = 3, y = 100$
- (c) $x = 4, y = 100$
- (d) $x = 3, y = 150$

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3. The mean proportional between 4 and 9 is:

- (a) 4 (b) 6
(c) 9 (d) 36

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4. What number must be added to each of the numbers 4, 6, 8, 11 in order to get the four numbers in proportion?

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5. Using componendo and dividendo solve for x .

$$\frac{\sqrt{2x+2} + \sqrt{2x-1}}{\sqrt{2x+1} - \sqrt{2x-1}} = 3$$

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6. If 6 is the mean proportion between two numbers x and y and 48 is the third proportional to x and y , find the numbers.

Comp 2023

7. The mean proportional between 9 and 16 is :

- (a) 25 (b) 144
(c) 7 (d) 12

Main 2022 Sem I

Solution

1.

Ans: (c) $M = 9, N = 4$

Explanation:

Since x is proportional to y , the ratio $\frac{x}{y}$ remains constant.

From the table:

$$\frac{6}{M} = \frac{12}{18} = \frac{N}{6}$$

Now,

$$\frac{12}{18} = \frac{2}{3}$$

So,

$$\frac{6}{M} = \frac{2}{3} \Rightarrow M = 9$$

and

$$\frac{N}{6} = \frac{2}{3} \Rightarrow N = 4$$

Hence, the correct answer is **(c)**.

2.

Ans: (d) $x = 3, y = 150$

Explanation:

Since the train moves with uniform speed, the ratio of distance to time remains constant.

$$\frac{60}{2} = \frac{90}{x} = \frac{y}{5}$$

From $\frac{60}{2} = 30$, we get:

$$\frac{90}{x} = 30 \Rightarrow x = 3 \text{ seconds}$$

Also,

$$\frac{y}{5} = 30 \Rightarrow y = 150 \text{ m}$$

Hence, the correct answer is **(d)**.

3.

Ans: (b) 6

Explanation:

Let the mean proportional between 4 and 9 be x .

Then, $4 : x = x : 9$

$$x^2 = 4 \times 9 = 36$$

$$x = \sqrt{36} = 6$$

Hence, the correct answer is **6**.

4.

Ans:

Let the number to be added be x .

According to the question, the four numbers will be in proportion:

$$(4 + x) : (6 + x) :: (8 + x) : (11 + x)$$

$$\frac{4 + x}{6 + x} = \frac{8 + x}{11 + x}$$

Cross-multiplying:

$$(4 + x)(11 + x) = (8 + x)(6 + x)$$

$$x^2 + 15x + 44 = x^2 + 14x + 48$$

$$x = 4$$

Hence, the required number to be added is 4.

5.

Ans:

Given:

$$\frac{\sqrt{2x+2} + \sqrt{2x-1}}{\sqrt{2x+2} - \sqrt{2x-1}} = 3$$

Using componendo and dividendo,

$$\frac{(\sqrt{2x+2} + \sqrt{2x-1}) + (\sqrt{2x+2} - \sqrt{2x-1})}{(\sqrt{2x+2} + \sqrt{2x-1}) - (\sqrt{2x+2} - \sqrt{2x-1})} = \frac{3+1}{3-1}$$

$$\frac{2\sqrt{2x+2}}{2\sqrt{2x-1}} = 2$$

$$\frac{\sqrt{2x+2}}{\sqrt{2x-1}} = 2$$

Squaring both sides,

$$\frac{2x+2}{2x-1} = 4$$

$$2x+2 = 4(2x-1)$$

$$2x+2 = 8x-4$$

$$6x = 6$$

$$x = 1$$

6.

Ans:

Since 6 is the mean proportion between x and y ,

$$\frac{x}{6} = \frac{6}{y}$$

$$xy = 36$$

$$x = \frac{36}{y} \dots\dots (1)$$

Since 48 is the third proportional to x and y ,

$$\frac{x}{y} = \frac{y}{48}$$

$$y^2 = 48x$$

Substituting the value of x from (1),

$$y^2 = 48 \times \frac{36}{y}$$

$$y^3 = 48 \times 36$$

$$y^3 = 12^3$$

$$y = 12$$

Substituting $y = 12$ in (1),

$$x = \frac{36}{12} = 3$$

Hence, the numbers are

$x = 3 \text{ and } y = 12$

7.

Ans: (d) 12

Explanation:

Let the mean proportional between 9 and 16 be x .

$$\frac{9}{x} = \frac{x}{16}$$

$$x^2 = 9 \times 16$$

$$x^2 = 144$$

$$x = \sqrt{144} = 12$$

Hence, the required mean proportional is **12**.

Thus, **(d)** is the correct option.